



PEM 2025 Mathematics Advanced Trial HSC Marking Guidelines

Mathematics Advanced HSC (Australian Islamic College of Sydney)



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PEM 2025 Mathematics Advanced

TRIAL HSC EXAMINATION SUGGESTED SOLUTIONS

Section I Multiple Choice Answer Key

Question	Answer	Solution
1	B	Distribution is negatively skewed, majority of scores are between 40-60, hence the box will be smaller on the left side
2	A	$E(X) = \frac{1}{6} \times \$36 + \frac{5}{6} \times \6 $E(X) = \$11$
3	C	$x - a = 10^3$ $x = 1000 + a$
4	A	Using two points i.e. (5,3) and (20,16) $m = \frac{16-3}{20-5}$ $m = \frac{13}{15}$ $3 = 5 \times \frac{13}{15} + b$ $b = -\frac{4}{3}$ $\therefore y = -\frac{4}{3} + \frac{13}{15}x$ $\therefore \text{ice cream} = -1.333 + 0.867 \times \text{temperature}$
5	D	$a = p$ $r = \frac{q}{p}$ $S_{\infty} = \frac{p}{1 - \frac{q}{p}}$ $S_{\infty} = \frac{p^2}{p - q}$

6	C	$\sin \theta = \frac{2}{5}, \therefore \cos \theta = \frac{\sqrt{21}}{5}$ $\sin(90^\circ - \theta) + \sin(180^\circ + \theta) = \cos \theta - \sin \theta$ $= \frac{\sqrt{21}}{5} - \frac{2}{5}$ $= \frac{\sqrt{21} - 2}{5}$
7	C	$\int_2^6 f(x) dx = \int_2^8 f(x) dx - \int_6^8 f(x) dx$ $\int_2^6 f(x) dx = C - D$
8	C	Length of centres of $C_1C_2 = \sqrt{(5-2)^2 + (-2-2)^2}$ $C_1C_2 = 5$ Considering the radii of both circles $r_1 + r_2 = 3 + 2$ $\therefore$ length of the centres is equal to the sum of the radii $\therefore$ both circles touch once externally
9	D	$f(x) = \tan(g(x^2))$ $f'(x) = \sec^2(g(x^2)) \times \frac{d}{dx}(g(x^2))$ $\frac{d}{dx}(g(x^2)) = 2xg'(x^2)$ $\therefore f'(x) = 2xg'(x^2)\sec^2(g(x^2))$

10	A	$f(g(x)) = \ln\left(\cos x + \frac{\sqrt{3}}{2}\right)$ $g(f(x)) = \cos\left(\ln\left(x + \frac{\sqrt{3}}{2}\right)\right)$ for $f(g(x))$ to be defined $\cos x + \frac{\sqrt{3}}{2} > 0$ $\therefore \cos x > -\frac{\sqrt{3}}{2}$ $x \in \left(-\frac{5\pi}{6}, \frac{5\pi}{6}\right) \cup \left(-\frac{17\pi}{6}, -\frac{7\pi}{6}\right) \cup \dots$ for $g(f(x))$ to be defined, $-\infty < \ln\left(x + \frac{\sqrt{3}}{2}\right) < \pi$ $0 < x + \frac{\sqrt{3}}{2} < e^\pi$ $\therefore -\frac{\sqrt{3}}{2} < x < e^\pi - \frac{\sqrt{3}}{2}$ Combining both conditions $x \in \left(-\frac{\sqrt{3}}{2}, \frac{5\pi}{6}\right)$
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Section II

Question 11 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$T_1 = 4, T_2 = 11, T_3 = 18$$

Question 11 (b)

Criteria	Marks
Provides correct solution	2
Attempts to use summation formula, or equivalent merit	1

Sample Answer

$$a = 4, d = 7, n = 100$$

$$S_{100} = \frac{100}{2}(2(4) + (100 - 1)(7))$$

$$S_{100} = 35050$$

Question 12

Criteria	Marks
Provides correct solution	2
Attempts to use area formula, or equivalent merit	1

Sample Answer

$$A = \frac{1}{2}(15)(20)\sin 124^\circ + \frac{1}{2}(18)(24)\sin 107^\circ$$

$$A = 330.9174632 \text{ cm}^2$$

$$A = 330.92 \text{ cm}^2 (2 \text{ d.p.})$$

Question 13 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$20 = 2x + 2y$$

$$2y = 20 - 2x$$

$$y = 10 - x$$

$$A = xy$$

$$A = x(10 - x)$$

$$A = 10x - x^2$$

Question 13 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$x = 2 \text{ or } x = 8$$

Question 13 (c)

Criteria	Marks
Provides correct solution	2
Finds correct maximum	1

Sample Answer

Max area = 25 cm^2 occurs at $x = 5 \text{ cm}$

Dimensions are $5 \text{ cm} \times 5 \text{ cm}$

Question 14

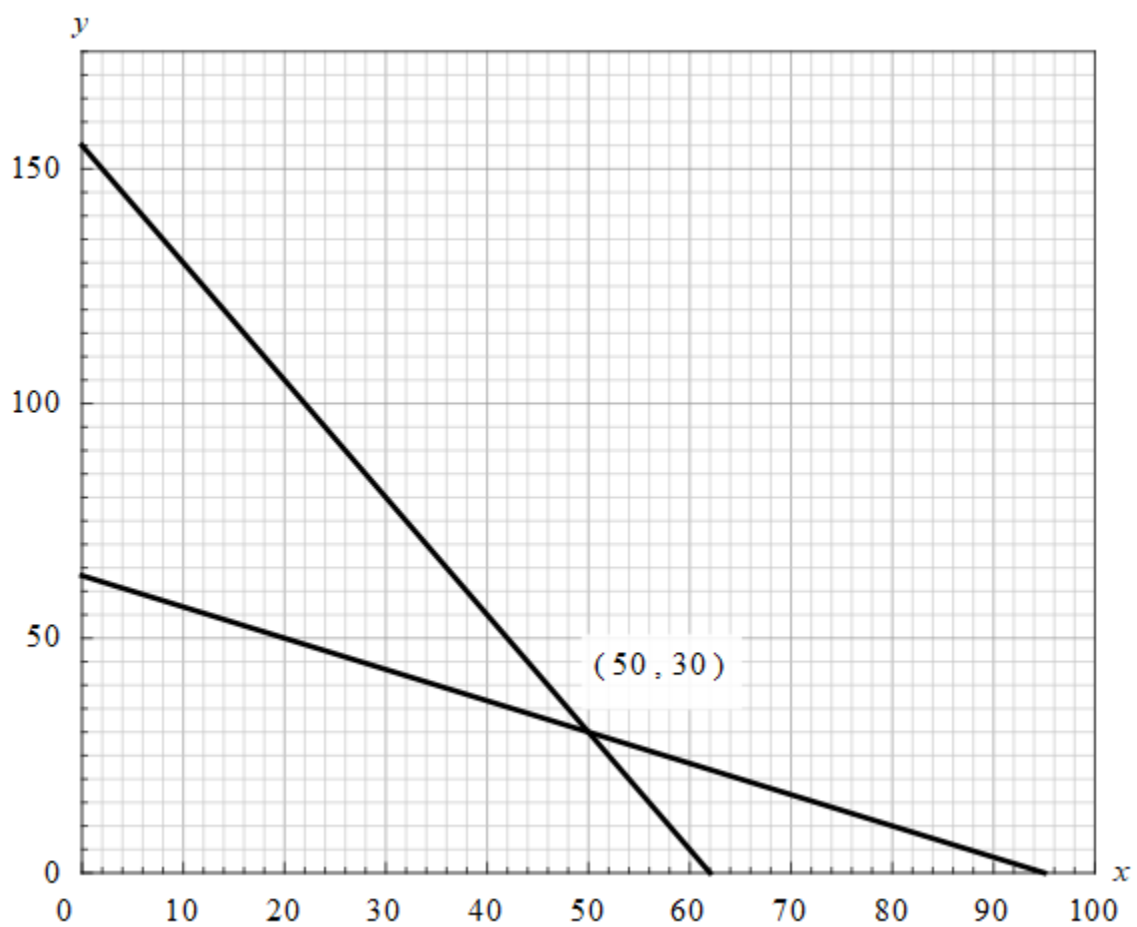
Criteria	Marks
Provides correct solution	3
Draws correct graph	2
Provides correct equation, or equivalent merit	1

Sample Answer

$$5x + 2y = 310$$

y – intercept is 155

x – intercept is 62



$\therefore$ adult ticket = \$50

$\therefore$ student ticket = \$30

Question 15 (a)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$C = \frac{k}{n}$$

$$k = 100 \times 75$$

$$\therefore C = \frac{7500}{n}$$

Question 15 (b)

Criteria	Marks
• Provides correct solution	1

Sample Answer

Many possible answers, some may include:

A bus will have a maximum capacity so large values of n are unreasonable

n must be a positive integer

Question 15 (c)

Criteria	Marks
• Provides correct solution	1

Sample Answer

$$80 = \frac{7500}{n}$$

$$n = \frac{7500}{80}$$

$$n = 93.75$$

But n must be an integer, $\therefore$ Cost per person cannot be \$80

Question 16 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$\angle DOP = (51 + 57) \times \frac{\pi}{180}$$

$$\angle DOP = \frac{3\pi}{5}$$

$$\angle DOP = 1.884955592...^c$$

Question 16 (b)

Criteria	Marks
Provides correct solution	2
Attempts to use arc length formula, or equivalent merit	1

Sample Answer

$$\text{Arc length} = (1740) \left(\frac{3\pi}{5} \right)$$

$$= 3279.82273... \text{ km}$$

$$= 3280 \text{ km}$$

Question 17

Criteria	Marks
• Provides correct solution	3
• Calculates one probability correctly, or equivalent merit	2
• Finds both areas, or equivalent merit	1

Sample Answer

$$\text{Total area} = 144\pi$$

$$\text{Shaded area} = 63\pi$$

$$\therefore P(\text{Shaded}) = \frac{7}{16}$$

$$P(\text{Both shaded}) = \left(\frac{7}{16}\right)^2$$

$$P(\text{Both shaded}) = \frac{49}{256}$$

$$P(\text{Unshaded and shaded}) = 2\left(\frac{7}{16}\right)\left(\frac{9}{16}\right)$$

$$P(\text{Unshaded and shaded}) = \frac{63}{128}$$

$$\frac{49}{256} < \frac{63}{128}$$

$\therefore$ Higher chance to have 1 dart land in each

Question 18

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

At $x = 1$, $y = -2$

$$m = 3x^2 - 3$$

$$m(1) = 0$$

Equation of tangent : $y - (-2) = 0(x - 1)$

$$\therefore y = -2$$

Question 19

Criteria	Marks
Provides correct solution	2
Attempts to use function of a function rule, or equivalent merit	1

Sample Answer

$$\int \frac{4x}{e^{x^2}} dx = \int 4xe^{-x^2} dx$$

$$= -2 \int -2xe^{-x^2} dx$$

$$= -2e^{-x^2} + C$$

Question 20

Criteria	Marks
• Provides correct solution	2
• Uses one correct trigonometric identity, or equivalent merit	1

Sample Answer

$$\begin{aligned}RHS : \frac{\operatorname{cosec} A}{\operatorname{cosec} A - \sin A} &= \frac{\frac{1}{\sin A}}{\frac{1}{\sin A} - \sin A} \times \frac{\sin A}{\sin A} \\&= \frac{1}{1 - \sin^2 A} \\&= \frac{1}{\cos^2 A} \\&= \sec^2 A \\&= LHS\end{aligned}$$

Question 21 (a)

Criteria	Marks
• Provides correct solution	2
• Finds one correct value	1

Sample Answer

$$A = \$236\,420.78 \times \frac{0.048}{12}$$

$$A = \$945.68312$$

$$A = \$945.68$$

$$B = \$236\,420.78 + \$945.68 - \$2\,741.05$$

$$B = \$234\,625.41$$

Question 21 (b)

Criteria	Marks
• Provides correct solution	2
• Finds total amount paid, or equivalent merit	1

Sample Answer

$$\text{Total paid} = 108 \times \$2\,741.05$$

$$\text{Total paid} = \$296\,033.40$$

$$\text{Interest} = \$296\,033.40 - \$240\,000$$

$$\text{Interest} = \$56\,033.40$$

Question 22

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$\int_{-2}^2 f(x) dx \approx (1 \times 6) + (1 \times 2) + (1 \times -2) + (1 \times -4)$$

$$\int_{-2}^2 f(x) dx \approx 2$$

Question 23

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$y = \ln\left(\frac{x^2}{\sqrt{x+1}}\right)$$

$$y = \ln(x^2) - \ln(\sqrt{x+1})$$

$$y = 2 \ln x - \frac{1}{2} \ln(x+1)$$

$$y' = \frac{2}{x} - \frac{1}{2(x+1)}$$

$$y' = \frac{3x+4}{2x(x+1)}$$

Or

$$y = \ln\left(\frac{x^2}{\sqrt{x+1}}\right)$$

$$y' = \frac{\frac{2x\sqrt{x+1} - \frac{x^2}{2\sqrt{x+1}}}{x+1}}{\frac{x^2}{\sqrt{x+1}}}$$

$$y' = \frac{2x(x+1) + \frac{x^2}{2(x+1)}}{x^2}$$

$$y' = \frac{\frac{2x(x+1)^2 + x^2}{2(x+1)}}{x^2}$$

$$y' = \frac{3x+4}{2x(x+1)}$$

Question 24 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$\begin{aligned}
 \frac{x+7}{x-1} &= \frac{x-1+8}{x-1} \\
 &= \frac{x-1}{x-1} + \frac{8}{x-1} \\
 &= 1 + \frac{8}{x-1}
 \end{aligned}$$

Question 24 (b)

Criteria	Marks
Provides correct solution	2
Correctly integration	1

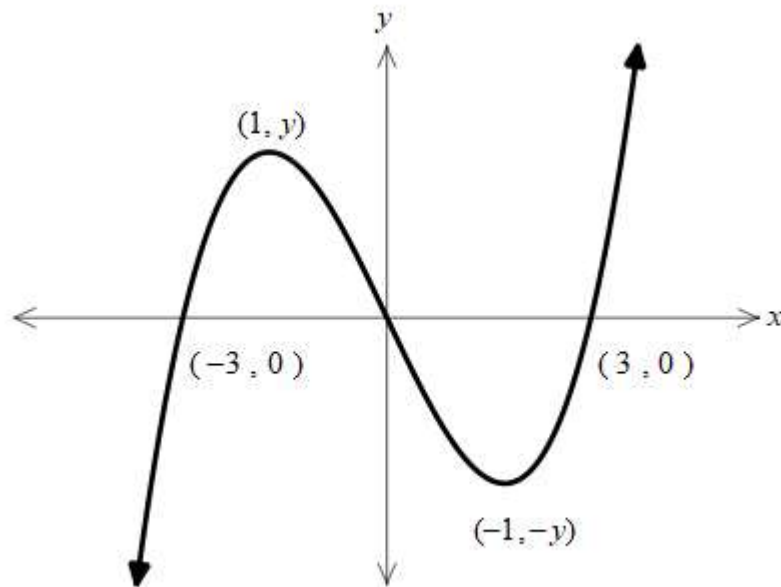
Sample Answer

$$\begin{aligned}
 \int_2^e \frac{x+7}{x-1} dx &= \int_2^e 1 + \frac{8}{x-1} dx \\
 &= \left[x + 8 \ln(x-1) \right]_2^e \\
 &= (e + 8 \ln(e-1)) - (2 + 8 \ln(2-1)) \\
 &= e + 8 \ln(e-1) - 2
 \end{aligned}$$

Question 25

Criteria	Marks
Provides correct solution	2
Graph has two correct properties	1

Sample Answer



Question 26 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$\frac{7+8+k}{7+8+k+5+3+1} = \frac{3}{4}$$

$$\frac{15+k}{24+k} = \frac{3}{4}$$

$$60+4k = 72+3k$$

$$\therefore k = 12$$

Question 26 (b)

Criteria	Marks
Provides correct solution	3
Calculates both mean and standard deviation	2
Calculates correct mean or standard deviation	1

Sample Answer

$$\mu = 1.77778...$$

$$\mu = 1.78 \text{ (2d.p.)}$$

$$\sigma = 1.311729744...$$

$$\sigma = 1.31$$

Within 1 standard deviation of the mean = $\mu \pm \sigma$

$\therefore$ Looking at scores between $(\mu - \sigma, \mu + \sigma)$

i.e. between (0.47 and 3.09)

$$= 8 + 12 + 5$$

$\therefore$ 25 families are within one standard deviation of the mean

Question 27 (a)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

Considering the standard scores of Chloe and Tiffany

$$\begin{cases} -0.5 = \frac{42 - \mu}{\sigma} \\ 1 = \frac{60 - \mu}{\sigma} \end{cases}$$

$$\frac{42 - \mu}{-0.5} = 60 - \mu$$

$$\mu = 48$$

$$\sigma = 12$$

Question 27 (b)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

Missing test mark is same as the mean mark

∴ Mean remains unchanged

Number of students increases by 1

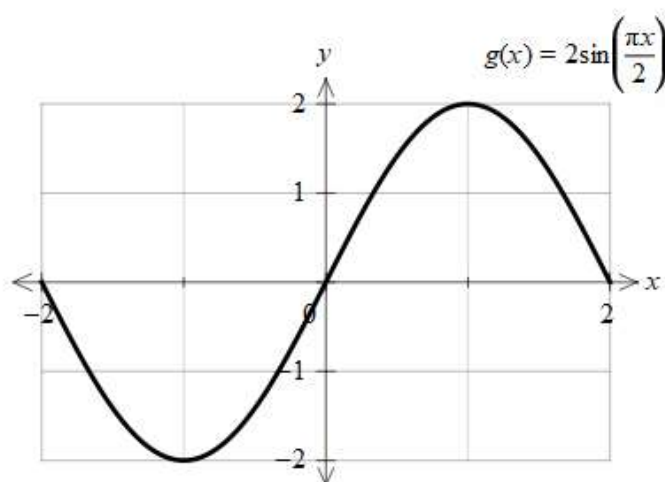
∴ Standard deviation decreases

∴ Chloe decreases, while Tiffany increases

Question 28 (a)

Criteria	Marks
Provides correct solution	2
Graph has two correct properties	1

Sample Answer



Question 28 (b)

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

$$A = \left| \int_{-1}^0 2 \sin \frac{\pi x}{2} dx \right| + \int_0^1 2 \sin \frac{\pi x}{2} dx$$

$$A = 2 \int_0^1 2 \sin \frac{\pi x}{2} dx$$

$$A = 4 \left[-\frac{2}{\pi} \cos \frac{\pi x}{2} \right]_0^1$$

$$A = 4 \left[\left(-\frac{2}{\pi} \cos \frac{2\pi}{2} \right) - \left(-\frac{2}{\pi} \cos 0 \right) \right]$$

$$A = 4 \left[\frac{2}{\pi} - \left(-\frac{2}{\pi} \right) \right]$$

$$A = \frac{16}{\pi} u^2$$

Question 29 (a)

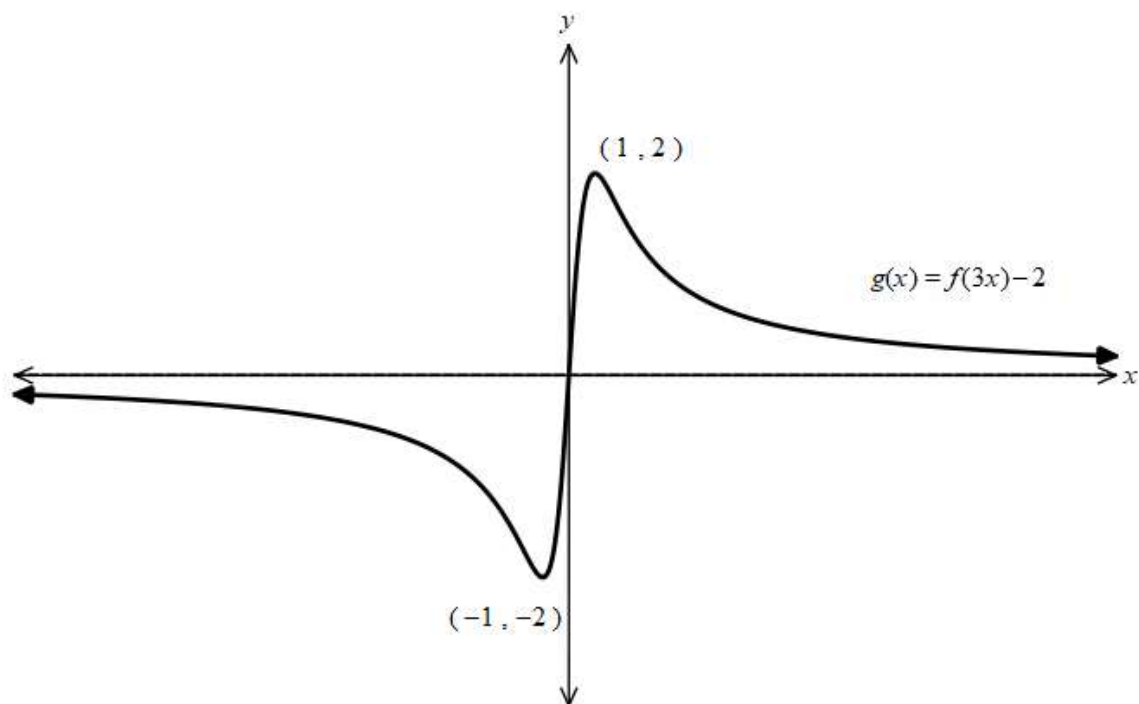
Criteria	Marks
Provides correct solution	1

Sample Answer

$$y \in [0, 2]$$

Question 29 (b)

Criteria	Marks
Provides correct solution	2
Graph has two correct properties	1

Sample Answer

Question 30 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$x = 1 - 2 \sin \left(2(0) + \frac{\pi}{6} \right)$$

$$x = 1 - 2 \left(\frac{1}{2} \right)$$

$$x = 0$$

Question 30 (b)

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

$$v = -4 \cos \left(2t + \frac{\pi}{6} \right)$$

$$v = 0$$

$$\cos \left(2t + \frac{\pi}{6} \right) = 0$$

$$2t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$2t = \frac{\pi}{3}, \frac{4\pi}{3}, \dots$$

$$t = \frac{\pi}{6}, \frac{2\pi}{3}, \dots$$

$$x = 1 - 2 \sin \left(2 \left(\frac{\pi}{6} \right) + \frac{\pi}{6} \right)$$

$$x = -1$$

$\therefore$ at rest after $\frac{\pi}{6}$ s, 1m to the left of the origin.

Question 30 (c)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$a = 8 \sin \left(2t + \frac{\pi}{6} \right)$$

$$a = 8 \sin \left(2 \left(\frac{\pi}{12} \right) + \frac{\pi}{6} \right)$$

$$a = 4\sqrt{3} \text{ ms}^{-2}$$

Question 31 (a)

Criteria	Marks
Provides correct solution	3
Attempts to determine the nature of the stationary points	2
Finds $f'(x)$ and attempts to solve $f'(x) = 0$	1

Sample Answer

$$y' = 4x^3 - 4x$$

$$y' = 0 \therefore 4x(x^2 - 1) = 0$$

$$x = -1, 0, 1$$

x	-2	-1	-0.5	0	0.5	1	2
y'	-24	0	1.5	0	-1.5	0	24

$$y(-1) = -1$$

$$y(0) = 0$$

$$y(1) = -1$$

$\therefore (-1, -1)$ is a minimum S.P.

$\therefore (0, 0)$ is a maximum S.P.

$\therefore (1, -1)$ is a minimum S.P.

Question 31 (b)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$y'' = 12x^2 - 4$$

$$y'' = 0 \therefore x^2 = \frac{1}{3}$$

$$x = \pm \sqrt{\frac{1}{3}}$$

x	-1	$-\sqrt{\frac{1}{3}}$	0	$\sqrt{\frac{1}{3}}$	1
y''	8	0	-4	0	8

$$y\left(\sqrt{\frac{1}{3}}\right) = -\frac{5}{9}$$

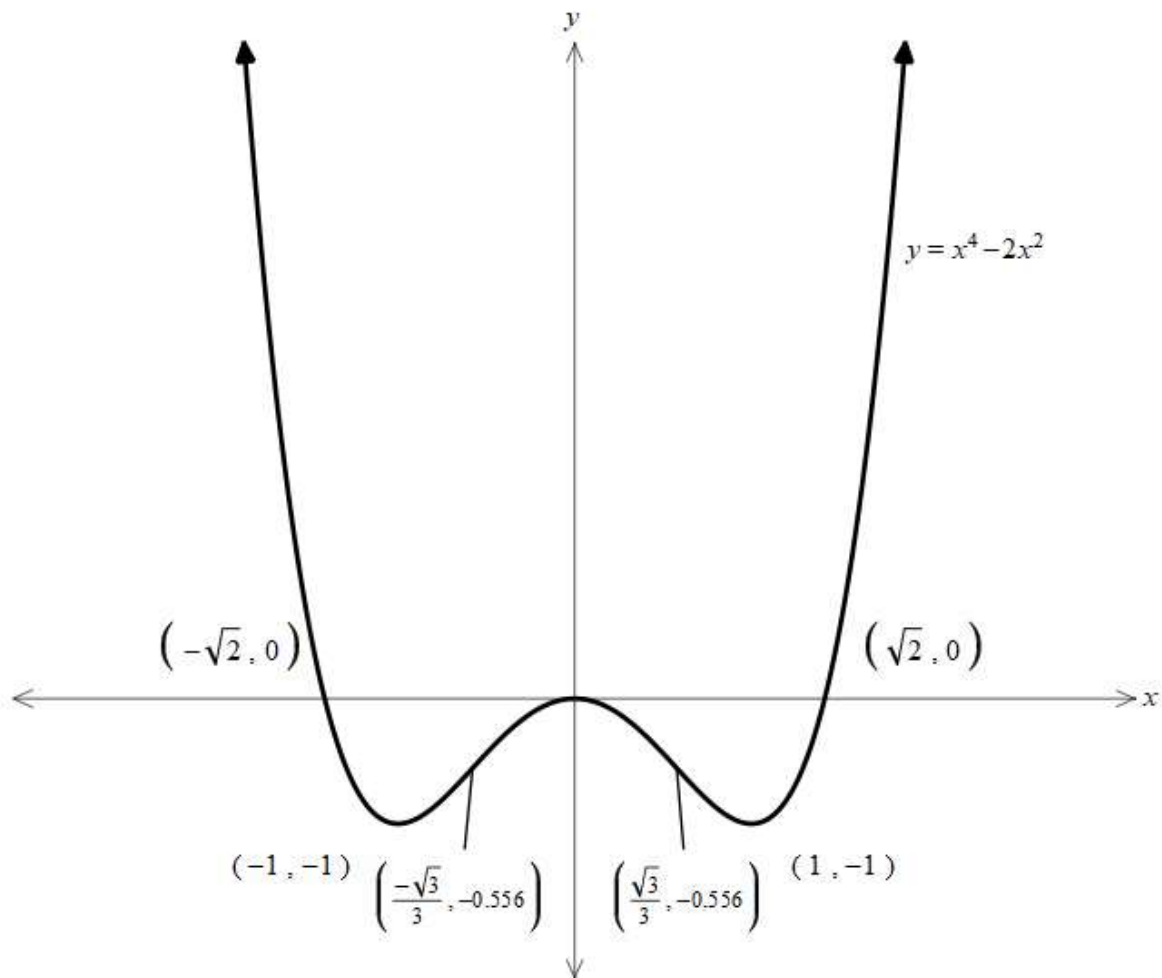
$$y\left(-\sqrt{\frac{1}{3}}\right) = -\frac{5}{9}$$

$$\therefore \left(-\sqrt{\frac{1}{3}}, -\frac{5}{9}\right) \text{ and } \left(\sqrt{\frac{1}{3}}, -\frac{5}{9}\right) \text{ are both points of inflexion}$$

Question 31 (c)

Criteria	Marks
Provides correct solution, showing all features	2
Correctly plots and labels stationary points and points of inflexion, or equivalent merit	1

Sample Answer



Question 32

Criteria	Marks
• Provides correct solution	4
• Sets up correct definite integral	3
• Finds equation of the quadratic	2
• Attempts to set up equation to find B , or equivalent merit	1

Sample Answer

$$f(-x) = f(x+3)$$

$$18 + B(-x) - (-x^2) = 18 + B(x+3) - (x+3)^2$$

$$-Bx - x^2 = Bx + 3B - x^2 - 6x + 9$$

$$0 = 2Bx - 6x + 3B - 9$$

$$0 = 2(B-3)x + 3(B-3)$$

$$B = 3$$

$$f(x) = 18 + 3x - x^2$$

$$f(x) = (6-x)(3+x)$$

$$A = \int_{-3}^6 18 + 3x - x^2 \, dx$$

$$A = \left[18x + \frac{3x^2}{2} - \frac{x^3}{3} \right]_{-3}^6$$

$$A = \left(18(6) + \frac{3(6)^2}{2} - \frac{(6)^3}{3} \right) - \left(18(3) + \frac{3(-3)^2}{2} - \frac{(-3)^3}{3} \right)$$

$$A = \frac{243}{2} u^2$$

Question 33 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$B(1)^2(4-(1)^2)=1$$

$$B(3)=1$$

$$B=\frac{1}{3}$$

Question 33 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$1-\left(\frac{1}{3}(0.5)^2(4-(0.5)^2)\right)=\frac{11}{16}$$

Question 33 (c)

Criteria	Marks
Provides correct solution	3
Finds both mode and standard deviation and attempts to use skewness formula, or equivalent merit	2
Finds the mode or standard deviation, or equivalent merit	1

Sample Answer

$$F(x) = \frac{1}{3}4x^2 - \frac{1}{3}x^4$$

$$\text{p.d.f} = \begin{cases} f(x) = \frac{8}{3}x - \frac{4}{3}x^3 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f'(x) = \frac{8}{3} - 4x^2$$

$$f'(x) = 0$$

$$\therefore x^2 = \frac{2}{3}$$

$$\therefore x = \sqrt{\frac{2}{3}}, (x > 0)$$

$$f''(x) = -8x$$

$$f''\left(\sqrt{\frac{2}{3}}\right) = -8\sqrt{\frac{2}{3}} < 0 \therefore \text{max}$$

$$\therefore \text{mode} = \sqrt{\frac{2}{3}}$$

$$\text{Var}(X) = \frac{4}{9} - \left(\frac{28}{45}\right)^2$$

$$\text{Var}(X) = \frac{116}{2025}$$

$$\sigma = \sqrt{\frac{116}{2025}}$$

$$\therefore \text{skewness coefficient} = \frac{\frac{28}{45} - \sqrt{\frac{2}{3}}}{\sqrt{\frac{116}{2025}}}$$

$$= -0.8117064616...$$

The distribution is negatively skewed

Question 34 (a)

Criteria	Marks
Provides correct solution	3
Attempts to use cosine rule or equivalent merit	2
Finds length of HC or equivalent merit	1

Sample Answer

Consider $\triangle AHC$

$$\tan 60^\circ = \frac{HC}{AC}$$

$$HC = x\sqrt{3} \text{ km}$$

Consider $\triangle BHC$

$$\tan 30^\circ = \frac{HC}{BC}$$

$$BC = \frac{x\sqrt{3}}{\frac{1}{\sqrt{3}}}$$

$$BC = 3x \text{ km}$$

In $\triangle ABC$

$$\angle ABC = 250^\circ - 130^\circ$$

$$\angle ABC = 120^\circ$$

$$(100\sqrt{13})^2 = x^2 + (3x)^2 - 2(x)(3x)\cos 120^\circ$$

$$130000 = 13x^2$$

$$x = 100 \text{ (} x > 0 \text{)}$$

$$\therefore AC = 100 \text{ km}$$

Question 34 (b)

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

Let A_1 and B_1 be the positions of the two battleships t hours after 9 a.m. respectively

$$A_1C = (100 - 20t) \text{ km and } B_1C = (300 - 50t) \text{ km}$$

In $\triangle A_1CB_1$

$$A_1B_1^2 = (100 - 20t)^2 + (300 - 50t)^2 - 2(100 - 20t)(300 - 50t)\cos 120^\circ$$

$$A_1B_1^2 = 10000 - 4000t + 400t^2 + 90000 - 30000t + 2500t^2 + 30000 - 5000t - 6000t + 1000t^2$$

$$A_1B_1^2 = 130000 - 45000t + 3900t^2$$

$$A_1B_1^2 = 100(1300 - 450t + 39t^2)$$

$$A_1B_1 = 10\sqrt{1300 - 450t + 39t^2}$$

Question 34 (c)

Criteria	Marks
Provides correct solution	2
Attempts to solve resulting quadratic, or equivalent merit	1

Sample Answer

$$10\sqrt{1300 - 450t + 39t^2} < 40$$

$$39t^2 - 450t + 1300 < 16$$

$$39t^2 - 450t + 1284 < 0$$

$$\frac{450 - \sqrt{450^2 - 4(39)(1284)}}{2(39)} < t < \frac{450 + \sqrt{450^2 - 4(39)(1284)}}{2(39)}$$

$$5.16844 < t < 6.37002$$

i.e. "on alert" between 2:10 p.m. and 3:22 p.m.

Question 35 (a)

Criteria	Marks
• Provides correct solution	3
• Finds simplified expression for A_n , or equivalent merit	2
• Finds an expression for A_2 , or equivalent merit	1

Sample Answer

$$A_1 = 5000(1 + 0.005)$$

$$A_2 = (5000 \times (1.01) + A_1)(1.005)$$

$$A_2 = 5000(1.01)(1.005) + 5000(1.005)^2$$

$$A_3 = (5000 \times (1.01)^2 + A_2)(1.005)$$

$$A_3 = 5000(1.01)^2(1.005) + 5000(1.01)(1.005)^2 + 5000(1.005)^3$$

$$A_n = 5000(1.01)^{n-1}(1.005) + 5000(1.01)^{n-2}(1.005)^2 + \dots + 5000(1.005)^n$$

$$A_n = 5000 \left[(1.01)^{n-1}(1.005) + (1.01)^{n-2}(1.005)^2 + \dots + (1.005)^n \right]$$

$$GP \ a = 1.005^n, \ r = \frac{1.01}{1.005}, \ n = n$$

$$A_n = 5000 \left[\frac{1.005^n \left(\left(\frac{1.01}{1.005} \right)^n - 1 \right)}{\frac{1.01}{1.005} - 1} \right]$$

$$A_n = 1005000(1.01^n - 1.005^n)$$

Question 35 (b)

Criteria	Marks
• Provides correct solution	1

Sample Answer

$$A_n = 3000000(1 + 0.005)^{n+24}$$

$$A_n = 3381479.33(1.005)^n$$

Note either is accepted

Question 35 (c)

Criteria	Marks
Provides correct solution	3
Finds an expression for $\left(\frac{1.01}{1.005}\right)^n$	2
Constructs correct inequality	1

Sample Answer

$$1005000(1.01^n - 1.005^n) > 3000000(1 + 0.005)^{n+24} \times 30\%$$

$$1.01^n - 1.005^n > 1.009396815(1.005)^n$$

$$\left(\frac{1.01}{1.005}\right)^n > 2.009396815$$

$$n > \log_{\frac{1.01}{1.005}}(2.009396815)$$

$$n > 140.6133802\dots$$

$\therefore$ At the end of September 2036, Charlene will have enough for a deposit

2025 PEM Trial HSC Advanced Mathematics

Mapping Grid

Section 1

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	2
2	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3
3	1	MA-E1 Logarithms and Exponentials	MA11-6	3
4	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	3-4
5	1	MA-M1 Modelling Financial Situations	MA12-4	3-4
6	1	MA-T2 Trigonometric Functions and Identities	MA11-4	4-5
7	1	MA-C4 Integral Calculus	MA12-7	4-5
8	1	MA-F1 Working with Functions	MA11-1	4-6
9	1	MA-C2 Differential Calculus	MA12-6	5-6
10	1	MA-F1 Working with Functions MA-T2 Trigonometric Functions and Identities	MA11-1 MA11-2 MA11-6	6

Section II

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
11 (a)	1	MA-M1 Modelling Financial Situations	MA12-4	2
11 (b)	2	MA-M1 Modelling Financial Situations	MA12-4	2-3
12	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	3-4
13 (a)	1	MA-F2 Graphing Techniques	MA12-1	2-3
13 (b)	1	MA-F2 Graphing Techniques	MA12-1	2-3
13 (c)	2	MA-F2 Graphing Techniques	MA12-1	3-4
14	3	MA-F1 Working with Functions	MA11-2	3-4
15 (a)	2	MA-F1 Working with Functions	MA11-1	2-4
15 (b)	1	MA-F1 Working with Functions	MA11-2	3-4

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
15 (c)	1	MA-F1 Working with Functions	MA11-1	3-4
16 (a)	1	MA-T1 Trigonometry and Measure of Angles	MA11-3	3
16 (b)	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	3-4
17	3	MA-S1 Probability and Discrete Probability Distributions	MA11-7	4-5
18	2	MA-C3 Applications of Differentiation	MA12-6	3-4
19	2	MA-C4 Integral Calculus	MA12-7	3-5
20	2	MA-T2 Trigonometric Functions and Identities	MA11-4	3-4
21 (a)	2	MA-M1 Modelling Financial Situations	MA12-2	3-4
21 (b)	2	MA-M1 Modelling Financial Situations	MA12-2	3-4
22	2	MA-C4 Integral Calculus	MA12-7	4-5
23	2	MA-C2 Differential Calculus	MA12-6	4-5
24 (a)	1	MA-F1 Working with Functions	MA11-1	3
24 (b)	2	MA-C4 Integral Calculus	MA12-7	4-5
25	2	MA-F1 Working with Functions	MA11-1	3-5
26 (a)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-4
26 (b)	3	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	4-5
27 (a)	2	MA-S3 Random Variables	MA12-8	3-5
27 (b)	2	MA-S3 Random Variables	MA12-8	4-5
28 (a)	2	MA-T3 Trigonometric Functions and Graphs	MA12-5	3-5
28 (b)	2	MA-C4 Integral Calculus	MA12-7	3-5
29 (a)	1	MA-F2 Graphing Techniques	MA12-1	3-4
29 (b)	2	MA-F2 Graphing Techniques	MA12-1	3-5
30 (a)	1	MA-C3 Applications of Differentiation	MA12-6	3-4
30 (b)	2	MA-C3 Applications of Differentiation	MA12-6	3-5
30 (c)	1	MA-C3 Applications of Differentiation	MA12-6	3-5
31 (a)	3	MA-C3 Applications of Differentiation	MA12-6	4-5
31 (b)	2	MA-C3 Applications of Differentiation	MA12-6	4-5
31 (c)	2	MA-C3 Applications of Differentiation	MA12-6	4-5
32	4	MA-C4 Integral Calculus	MA12-7	5-6
33 (a)	1	MA-S3 Random Variables	MA12-8	4-5
33 (b)	1	MA-S3 Random Variables	MA12-8	3-4
33 (c)	3	MA-S3 Random Variables	MA12-8	4-6
34 (a)	3	MA-T1 Trigonometry and Measure of Angles	MA11-3	4-5
34 (b)	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	5-6

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
34 (c)	2	MA-F2 Graphing Techniques	MA12-1	5-6
35 (a)	3	MA-M1 Modelling Financial Situations	MA12-2	6
35 (b)	1	MA-M1 Modelling Financial Situations	MA12-2	3
35 (c)	3	MA-M1 Modelling Financial Situations	MA12-2	5-6